

Pedestrian Trajectory Prediction With Reinforcement Learning From a Human-Designed Reward Model

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Abstract—Accurately predicting pedestrian trajectories in safety-critical scenarios is crucial for autonomous vehicles. In such contexts, human behaviors, i.e., pedestrians’ responses when noticing and attempting to avoid dangerous vehicles, play a decisive role in determining safe outcomes. Most current pedestrian trajectory prediction models are trained on data collected under low-risk scenarios. Consequently, although they perform well in routine domains, their generalization to high-risk interactions involving human reactions remains limited due to both data bias and models’ structural inadequacies. This article proposes a fine-tuned transformer trajectory prediction model with reinforcement learning from a human-designed reward model under proximal policy optimization. This design effectively mitigates the performance degradation caused by data bias and enhances prediction capability for rare behaviors. To support model training and evaluation, we constructed a pedestrian trajectory dataset specifically designed for safety-critical scenarios. Comprehensive evaluations against several state-of-the-art models demonstrate that our model achieves average reduction of 8.8% in brierADE and 11.2% in brierFDE. A study based on human evaluation shows that predictions from the fine-tuned model better emphasizing rare behaviors. Overall, the proposed model maintains comparable performance on common behaviors while substantially improving the prediction of abrupt behavioral changes. This advancement enables autonomous vehicles to accurately anticipate pedestrian trajectories in safety-critical scenarios, thereby enhancing safety.

Introduction

Road traffic accidents remain a major global public health concern, causing approximately 1.3 million deaths each year [1]. As vulnerable road users, pedestrians are at high risk of injury in collisions with motor vehicles. Autonomous vehicles, equipped with advanced sensing and safety algorithms, have the potential to reduce pedestrian–vehicle collisions. Data-driven trajectory prediction models have achieved notable accuracy [2]. However, most existing methods neglect the long-tail distribution issue of pedestrian trajectories data [3]. In the real world, the majority of head samples represent common behaviors such as constant-speed motion, whereas rare tail samples involve abrupt changes in speed and direction. This severe imbalance hinders effective learning of tail trajectories. The problem is particularly acute in safety-critical traffic scenarios, where road users’ actions can immediately affect safety and may result in collisions [1]. In such situations, pedestrians may either maintain their motion or suddenly change behavior to avoid hazardous vehicles [Figure 1(a)], leading to an even more pronounced long-tail distribution. Improving long-tail trajectory prediction in safety-critical scenarios is therefore essential for intelligent vehicle safety decision making and remains a key bottleneck for the real-world deployment of autonomous vehicles.

Pedestrian trajectory prediction has improved rapidly, with numerous models proposed and the state-of-the-art (SOTA) scores continually updated on public datasets [4], [5]. Existing models typically rely on generative or Transformer-based architectures to learn the distribution of pedestrian behaviors and forecast multiple possible trajectories. However, datasets such as ETH/UCY are collected in campus [6], [7], and traffic datasets like Argoverse, Waymo, inD, and NuScenes

mainly represent normal traffic scenarios [8], [9], [10], [11]. As a result, the distribution of trajectories in these datasets differs substantially from that in safety-critical scenarios, limiting their direct application. Although the I See You dataset records 170 near-accident pedestrian–vehicle interaction cases, it does not include collision events [12]. The SAVeD dataset contains both near-miss and crash events. However, the limited number of pedestrian-related cases is insufficient to support effective model training [13].

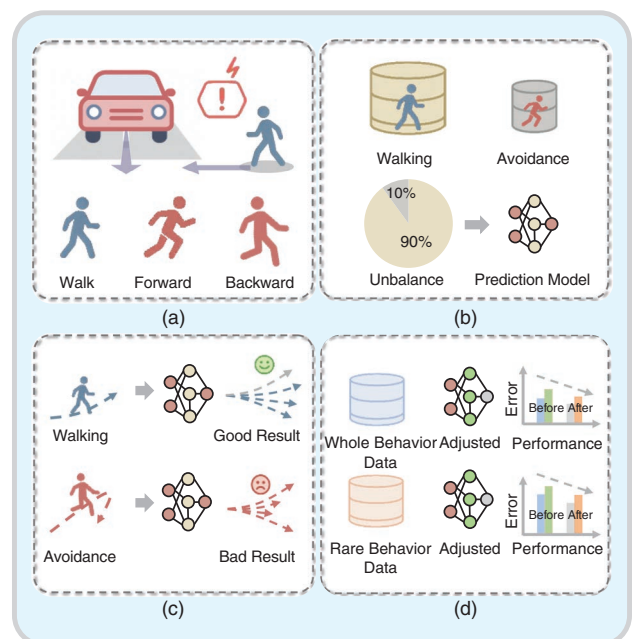


FIG 1 Problem description. (a) Pedestrian behaviors in safety-critical scenarios. (b) Unbalanced data between common behaviors and rare-avoidance behaviors. (c) Model collapse and imbalance for common and rare trajectory prediction. (d) Fine-tuned expectation.

In safety-critical scenarios, particularly those involving collisions, pedestrians often exhibit avoidance behaviors, such as backward avoidance, making sudden stops, or accelerating forward. These behaviors are closely related to the likelihood of subsequent collisions. During data collection, such rapid behavioral changes result in a pronounced long-tail distribution. Common behaviors are heavily overrepresented, while safety-critical avoidance actions remain extremely scarce [Figure 1(b)]. This severe imbalance causes existing models to assign disproportionately high probabilities to common behaviors and fail to capture rare but crucial avoidance actions, thereby limiting their effectiveness in high-risk scenarios [Figure 1(c)]. This issue has been commonly referred to as a *data imbalance problem* in prior studies [14]. Addressing this problem involves two key challenges: 1) constructing a dataset that accurately reflects pedestrian trajectory distributions in collision-related safety-critical scenarios and 2) designing a model architecture that can handle the data imbalance problem, preventing the model from overfitting to common behaviors while neglecting rare ones. Fortunately, virtual-reality (VR) technology makes it possible to collect pedestrian behaviors within virtual safety-critical scenarios [1]. In parallel, reinforcement learning from a human-designed reward model enables the model to align with human rules [14] and assign more attention to rare behaviors. Combining these techniques can enhance predictive performance and contributes to autonomous vehicles [Figure 1(d)].

To address the aforementioned challenge, this study proposes a fine-tuned pedestrian trajectory prediction model with human-designed reward model. We first pretrained a multimodal pedestrian trajectory prediction model using a Transformer architecture. Second, in rare-behavior cases where the pretrained model underperforms, we manually ranking predicted trajectories. This ranked feedback serves as an additional supervisory signal to guide model refinement. Third, under the proximal policy optimization (PPO) reinforcement learning framework, we used these human-designed signals to iteratively optimize the pretrained model, enhancing its ability to predict rare behaviors while preserving accuracy for common ones. To address the lack of trajectory datasets for collision-related safety-critical scenarios, we further employ VR technologies integrated with a pedestrian 3D motion simulator to construct immersive hazardous environments within CARLA. Volunteers are engaged in extensive interaction trials, enabling the construction of a safety-critical pedestrian trajectory dataset. Finally, we conduct a comprehensive evaluation of the proposed model using established trajectory prediction metrics, demonstrating consistent performance on common behaviors and improvements in predicting rare avoidance behaviors.

The contributions of this study are summarized as follows:

- A fine-tuning trajectory prediction framework based on a human-designed reward model is proposed, which integrates reinforcement learning with human supervisory signals to enhance model generalization across multimodal trajectories in safety-critical scenarios.
- We constructed a collision-involved safety-critical pedestrian trajectory prediction dataset. The dataset is derived from large-scale VR-based volunteer experiments, capturing diverse pedestrian-vehicle interactions under hazardous scenarios.
- We conducted extensive experimental validation of our pedestrian trajectory prediction model on the safety-critical dataset. The fine-tuned model outperforms several SOTA baselines in standard evaluation metrics and achieves good prediction accuracy for rare avoidance behaviors.

The structure of this article is as follows: First, the “Related Work” section reviews pedestrian trajectory prediction models and reinforcement learning from human feedback (RLHF) as well as datasets and the long-tail problem in a trajectory prediction task. Then, the problem statement and formulation of the proposed Transformer-based pedestrian trajectory prediction pretrained model are introduced in the “Trajectory Prediction Pre-Trained Model” section. The “Reinforcement Learning From a Human-Designed Reward Model” section provides a detailed description of reinforcement learning with a human-designed reward model. Finally, the training dataset for safety-critical scenarios is shown in the “Pedestrian Dataset for Safety-Critical Scenarios” section. Experimental design and result analysis are described in the “Experiment Design and Result Analysis” section, followed by conclusions in the “Conclusion” section.

Related Work

Pedestrian Trajectory Prediction Models

Pedestrian trajectory prediction has long been a popular research area, with various models proposed over time. These models can be classified into rule- and learning-based models [2]. Rule-based trajectory prediction models, such as the social force model and its variants, typically assume that pedestrian movement follows specific motion laws [15], [16]. These models describe pedestrian behaviors using explicit mathematical formulations, offering strong interpretability. However, due to the nonlinear nature of pedestrian trajectories, such models are often limited to specific scenarios, and their weak generalizability constrains their application.

Learning-based trajectory prediction models have demonstrated strong trajectory prediction capability. Recurrent neural networks {e.g., Social-LSTM (long short-term

memory) [17] and Group LSTM [18]} and convolution neural networks (CNNs) (e.g., Behavior-CNN [19] and Conv2D [20]) focus on the temporal and spatial features of pedestrian trajectories, respectively. To jointly capture both temporal and spatial characteristics, graph neural networks, such as Spares Graph Convolutional Network (SGCN) [21] and Spatio-Temporal Graph CNN (Social-STGCNN) [22], have been applied. Also, attention mechanisms, including graph attention networks, enable direct modeling of interactions among pedestrians [23], [24], [25], [26]. Furthermore, considering that pedestrians may execute multiple possible future behaviors, predicting multimodal trajectories is essential. To tackle this challenge, deep generative models, leveraging their ability to fit and sample from data distributions, can generate multiple potential trajectories together with their associated probabilities. Such methods have achieved strong performance in pedestrian trajectory prediction, exemplified by generative adversarial networks [27], [28], and conditional variational autoencoders [4], [29], [30]. In addition, Gaussian mixture models and tree-based models have also been employed to predict diverse trajectories [31], [32]. More recently, Transformer-based models have been introduced for sequence prediction tasks [33], [34], [35]. With their superior ability to capture temporal, spatial, and social features, models such as STAR, AgentFormer, and Trajectory Unified Transformer (TUTR) have achieved SOTA performance in pedestrian trajectory prediction [5], [36], [37]. In recent years, diffusion models have been widely adopted for multimodal pedestrian trajectory prediction due to their strong fitting capacity [39], [40], [41], [42], [43]. Moreover, with the rapid advancement of large language models (LLMs), whose superior contextual understanding and generation capabilities have inspired reformulating trajectory prediction as a question-answering task, language-based multimodal predictors have also achieved promising performance [44], [45], [46], [47].

For pedestrian trajectory prediction in safety-critical scenarios, limited by dataset availability, only a few studies have employed continuous Gaussian process inverse reinforcement learning to recover pedestrians' reward functions and infer avoidance mechanisms in conflict scenarios [48]. In addition, imitation learning was used to model human-vehicle interactions in virtual traffic environments [49]. This indicates that pedestrian trajectory prediction in safety-critical scenarios remains in its early stage. Given the strong modeling capacity of transformer-based models, we adopt them as a foundation for fine-tuning to enable effective deployment in safety-critical scenarios.

Datasets for Pedestrian Trajectory Prediction

Pedestrian trajectory prediction has been widely supported by large-scale public datasets. Among them, the ETH/UCY datasets are the most widely used benchmark, capturing diverse pedestrian interactions on university campuses

[6], [7]. However, such datasets lack information on pedestrian-vehicle interactions in traffic scenarios and thus cannot directly support multimodal pedestrian trajectory prediction in safety-critical scenarios. Datasets such as Argoverse, Waymo, inD, and NuScenes provide traffic data, including participants' trajectories and map information [8], [9], [10], [11]. Nevertheless, safety-critical scenarios follow a long-tail distribution, where normal cases dominate the data. By contrast, scenarios with abrupt behavioral changes are severely underrepresented. As a result, models trained on these datasets often suffer degraded performance when tested in safety-critical scenarios.

Due to the high risk in safety-critical scenarios, collecting pedestrian-vehicle interaction data remains challenging. The I See You dataset includes 170 real-world near-miss pedestrian-vehicle interactions but contains no collision cases [12]. The SAVeD dataset records 2,119 first-person videos capturing vehicle interactions with road users under risky conditions. However, pedestrians constitute only a small part of the data [13]. The IDD-CRS dataset collects critical road scenarios via dashcams and integrates both vehicle and pedestrian behaviors, yet it is not publicly available [50]. Similarly, JaywalkerVR acquires pedestrian-vehicle interaction data in safety-critical scenarios using a VR human-in-the-loop simulator, but the dataset is also not open sourced [51]. In summary, constructing a trajectory dataset that explicitly captures collision-related behaviors in safety-critical scenarios is essential for addressing trajectory prediction in such contexts.

Long-Tail Trajectory Prediction

The long-tail problem arises from imbalanced data distributions, where models overfit dominant samples while neglecting rare tail cases. Common ways include resampling and loss reweighting. However, undersampling may impair data representativeness [52], whereas oversampling often leads to overfitting [53]. Loss reweighting assigns higher importance to tail samples but can degrade performance on head data [54]. In pedestrian trajectory prediction, weighting samples by prediction error and incorporating contrastive learning have been shown to enhance performance on tail trajectories [55]. Moreover, combining loss reweighting with contrastive learning can further alleviate trajectory class imbalance [56]. Recently, dynamic subclass-balanced contrastive learning has been proposed, employing adaptive motion-pattern refinement to better identify rare motion modes [3]. As pedestrian trajectory prediction matures, addressing the long-tail problem has become a critical focus as rare yet safety-critical behaviors disproportionately impact real-world performance.

RLHF

With the development of LLMs, it has become essential to align model outputs with human preferences [57]. RLHF

has emerged as a key approach to address this need [58]. Its central idea is to construct a reward model from limited human feedback data, thereby avoiding the difficulty of explicitly designing a reward function. Such reward models encode human preferences, and reinforcement learning is then used to fine-tune model parameters. The model will be optimized toward human-designed reward while keeping its original performance. Therefore, the challenge lies in the effective design of limited datasets that represent human preferences. Fine-tuning methods include PPO, direct preference optimization (DPO), and group relative policy optimization (GRPO). PPO relies on a reward model within an explicit reinforcement learning framework to optimize model outputs toward human-aligned rewards [59]. DPO directly uses human preference comparison data to encourage the model to generate better outputs while avoiding worse ones [60]. GRPO stabilizes policy updates by leveraging relative reward comparisons within groups, thereby mitigating the instability caused by scale differences in multiagent or human feedbacks [61]. These methods have recently been applied in trajectory prediction tasks to generate human-aligned trajectories, providing an approach to enhancing models' ability to predict rare behaviors in safety-critical scenarios [14].

Trajectory Prediction Pretrained Model

This section provides a detailed description of the pretrained pedestrian trajectory prediction model, which serves as the baseline for subsequent fine-tuning.

Problem Statement

The objective of trajectory prediction is to forecast future trajectories based on the historical trajectories of the pedestrian and neighboring vehicles. To formalize this process, we assume that the total duration of an interaction segment is T , and that the scenario involves N traffic par-

ticipants (i.e., in our task, $N = 2$). $\{x_t, y_t\}_{t=1, n=1}^{T, N}$ denotes the ground-truth trajectories of the pedestrian and the vehicle at each time stamp.

The model inputs are the observed history of the road users, denoted as $\{x_t, y_t\}_{t=1, n=1}^{T_{\text{obs}}, N}$. The model output is the future trajectory of the pedestrian, defined as $\{x_t, y_t\}_{t=T_{\text{obs}}+1}^T$. As pedestrians may have multiple future behaviors, the model produces multiple predicted trajectories together with their associated probabilities.

Pretrained Model Architecture

The design of the prediction model is based on an SOTA pedestrian trajectory prediction model, i.e., TUTR [37], which employs a Transformer encoder-decoder architecture to predict future trajectories. To adapt the model for subsequent reinforcement learning fine-tuning, we modified the position of the classification prediction output and exchange the classification loss from cross-entropy loss to Kullback-Liebler (KL) divergence loss. A brief introduction to the modified trajectory prediction model is provided here:

- *Motion encoder*: For the standardized model training, the interaction data between pedestrians and vehicles are normalized through translation and rotation. Considering the multiple possible future behaviors of pedestrians, we adopt the k -means method to cluster pedestrians' future trajectories. The resulting cluster centers are defined as pedestrian future motion modes and formulated as

$$\min_{\{\mu_m\}_{m=1}^M, \{c_i\}_{i=1}^N} \sum_{i=1}^N \sum_{t=T_{\text{obs}}+1}^T [(x_{i,t} - \mu_{c_i,t}^x)^2 + (y_{i,t} - \mu_{c_i,t}^y)^2] \quad (1)$$

$c_i \in \{1, \dots, M\}$

where N denotes the number of trajectory samples, t is the time stamp, i is the trajectory index, and m represents the cluster index. M is the number of clusters, and μ_m is

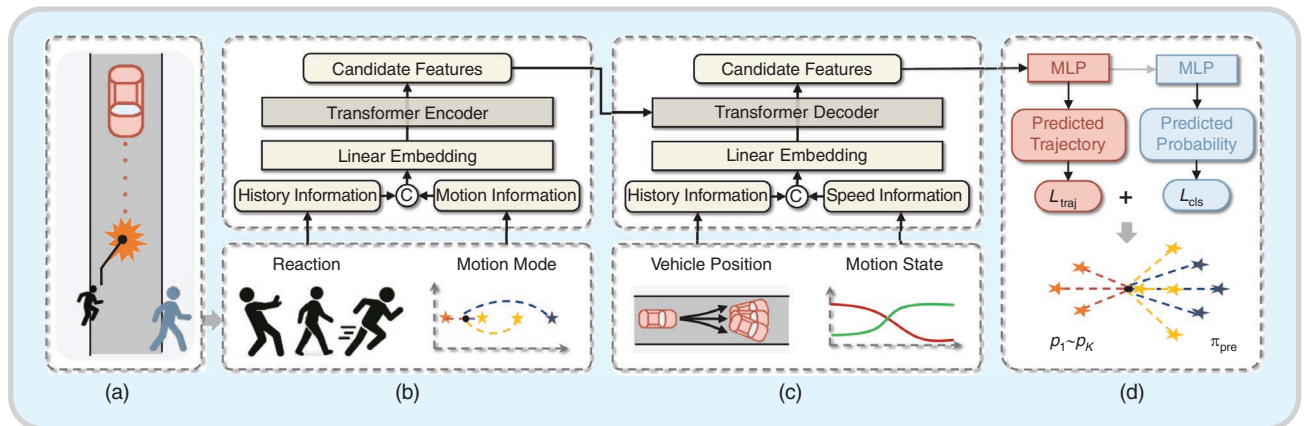


FIG 2 Pretrained trajectory prediction model architecture. (a) Safety-critical scenarios, where a jaywalking pedestrian collides with a vehicle. (b) Motion encoder that encodes pedestrians' historical trajectories and their future motion modes. (c) Social decoder, which receives the encoder output together with the historical trajectory and velocity of the neighboring vehicle. (d) Output module, which consists of a regression head and a classification head to generate the trajectories and probabilities. MLP: multilayer perceptron.

the cluster center trajectory. $x_{i,t}$ and $y_{i,t}$ are the ground-truth coordinates of the i th trajectory at t , while $\mu_{c_i,t}^x$ and $\mu_{c_i,t}^y$ denote the coordinates of the cluster center trajectory at t . c_i indicates the cluster assignment of the i trajectory.

The motion encoder first encodes the pedestrians' historical trajectory and future motion modes through a linear transformation, and then concatenates them to obtain a feature E_e . This feature is subsequently fed into a standard Transformer encoder. The features are further processed by a feedforward neural network with residual connections, yielding the output of the motion encoder [Figure 2(b)].

- **Social decoder:** The encoder outputs are used as the keys and values. The vehicle's historical trajectory and velocity are linearly transformed and taken as the queries, which are then fed into a standard Transformer decoder. Cross attention is employed to capture the interdependency between pedestrian and vehicle features. These features are then processed by a feedforward neural network with residual connections. The output serves as the final result of the decoder [Figure 2(c)].
- **Total loss:** Two parallel multilayer perceptrons (MLPs) are employed to process the decoder output E_{i,c_i} [Figure 2(d)]. And the regression head maps E_{i,c_i} to future trajectory coordinates. The classification head projects E_{i,c_i} onto the probability distribution over S candidate trajectories

$$\begin{aligned}\hat{Y}_{i,c_i} &= W_2 \sigma(W_1 E_{i,c_i} + b_1) + b_2, \\ \hat{s}_{i,c_i} &= W_c E_{i,c_i} + b_c, \\ \hat{p}_{i,c_i} &= \frac{\exp(\hat{s}_{i,c_i})}{\sum_{j=1}^S \exp(\hat{s}_{i,j})}\end{aligned}\quad (2)$$

where $\hat{Y}_{i,c_i}$ denotes the predicted coordinate sequence of the c_i th motion mode for the i th sample. $\sigma(\cdot)$ presents a nonlinear activation function, and W_1 , W_2 , W_c , b_1 , b_2 , b_c are the learnable parameters of the MLP. $\hat{s}_{i,c_i}$ and $\hat{p}_{i,c_i}$ correspond to the logit score and the probability, respectively.

The training loss combines the Smooth L1 Loss between predicted and ground-truth trajectories and the KL divergence loss between the predicted probability distribution and the soft-label distribution from trajectory the dataset, formulated as

$$\begin{aligned}d_{i,c_i} &= \sum_{t=T_{\text{obs}}+1}^T [(x_{i,t} - \mu_{c_i,t}^x)^2 + (y_{i,t} - \mu_{c_i,t}^y)^2], \\ q_{i,c_i} &= \frac{\exp(-d_{i,c_i})}{\sum_{j=1}^S \exp(-d_{i,j})}, \\ \mathcal{L}_{\text{reg}} &= \frac{1}{B} \sum_{b=1}^B \text{SmoothL1}(\hat{Y}_{i,k}, Y_{i,k}), \\ \mathcal{L}_{\text{cls}} &= \frac{1}{B} \sum_{b=1}^B \text{KL}(q_{i,c_i}, \hat{p}_{i,c_i}) \\ \mathcal{L}_{\text{Total}} &= \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{cls}}\end{aligned}\quad (3)$$

where, d_{i,c_i} denotes the distance between i th ground-truth trajectory and c_i th motion mode. q_{i,c_i} represents the constructed soft-label probability, where a motion mode closer to the ground truth can be assigned a higher probability. B is the batch size. $\mathcal{L}_{\text{reg}}$ refers to the regression loss, $\mathcal{L}_{\text{cls}}$ is the classification loss, and $\mathcal{L}_{\text{Total}}$ is the overall training loss.

In this way, a pedestrian trajectory pretrained model is constructed, which will be further fine-tuned to enable trajectory prediction in safety-critical scenarios.

Reinforcement Learning From a Human-Designed Reward Model

The goal of fine-tuning is to align model outputs with human rules, thereby adjusting the probability allocation between common and rare behaviors. However, directly training the policy from human scores is impractical. Therefore, we need to train a reward model on human-designed dataset to replace direct human scoring. We then fine-tune the policy with reinforcement learning to maximize the reward model's score while applying a KL constraint to keep the policy close to a reference policy. This section details the fine-tuning architecture and the overall training procedure.

Core Components of PPO

Actor Model

The actor model is the central trainable policy in the PPO loop. It has the same structure as the pretrained model, but the classification head is learnable. The purpose is to align predictions with human rules by maximizing the reward model's score while remaining close to a reference policy, thereby selectively emphasizing rare behaviors without degrading common behaviors. We initialize the actor from the pretrained checkpoint to preserve baseline competence.

Reward Model

Within the PPO, refining the outputs of a pretrained model to produce trajectories that align with human judgment relies on a reward model that can serve as a surrogate for human scores. Such a model should recognize the well-performing outputs of the pretrained model. It also needs to judge which outputs are better when prediction performance is poor. Therefore, a dataset that incorporates both types of aforementioned cases is required for training the reward model.

Human-Designed Dataset

In our task, the training set contains 15,286 trajectories, including common and rare behaviors. Eight motion modes were clustered from the training set: three constant-speed normal walking, forward acceleration, stopping, stopping followed by backward, normal walking followed by backward, and constant-speed backward. Accordingly, the pre-

trained model generates eight candidate trajectories for each observation. To identify rare behaviors, we adopt a top-four criterion based on a weighted combination of average displacement error (ADE) and final displacement error (FDE). If the trajectory closest to the ground truth ranks within the top four in probability, the ground-truth trajectory is classified as common behavior. Otherwise, it is regarded as rare behavior. Therefore, the training set is partitioned into 13,577 common trajectories and 1,709 rare trajectories.

The pretrained model performs poorly on rare-behavior trajectories, necessitating reinforcement learning to reallocate prediction probabilities. A key component is a reward model capable of identifying high-quality predictions and guiding optimization. Accordingly, the reward model must learn to rank the eight candidate trajectories generated by the pretrained model when its predictions are unreliable. To this end, human ranking is introduced. For rare-behavior cases, the eight candidate trajectories are reordered using a weighted combination of ADE and FDE. Based on this procedure, we construct one part of the human-designed dataset comprising 1,709 cases, each including pedestrian and vehicle historical trajectory and velocity; eight candidate trajectories generated by the pretrained model; and an objective ranking from zero to seven.

In addition, the reward model must recognize that the pretrained model produces reliable predictions for common behaviors. However, including all common-behavior predictions in the human-designed dataset would introduce new data imbalance. Treating common and rare behaviors as equally important, we sample 1,709 cases from the common-behavior trajectories. To maintain dataset representativeness, after sampling, it is necessary to keep the proportion of cases in which the probability of the trajectory closest to the ground truth appears within top one to top four consistent with that before sampling. In this way, we construct the other part of the human-designed dataset. The total dataset includes both high-quality outputs from the pretrained model and reranked low-quality predictions [Figure 3(a)].

Reward Model Architecture

The reward model also adopts the Transformer architecture. In contrast to the pretrained model, it modifies the encoder input and the loss function. Specifically, the pedestrian's historical trajectory and the candidate trajectories generated by the pretrained model are encoded and concatenated as the input to the Transformer encoder. Similarly, the historical trajectories and velocities of hazardous vehicles are transformed and concatenated to form the queries, and the encoder outputs serve as keys and values for the Transformer decoder. The features produced by the decoder are then processed by an MLP to yield a scalar score for each candidate trajectory [Figure 3(b)–(d)].

To enable the reward model to distinguish between better and worse trajectories, we introduce the ListNet loss [62], which learns candidate trajectories' relative rankings. It normalizes both the ranking labels of the candidate trajectories and the scores from the reward model into probability distributions and minimizes their difference using cross entropy. The specific formulation is as follows:

$$P_l^{\text{gt}} = \frac{\exp(-r_l)}{\sum_{k=1}^K \exp(-r_k)},$$

$$P_l^{\text{pred}} = \frac{\exp(c_l)}{\sum_{k=1}^K \exp(c_k)},$$

$$\mathcal{L}_{\text{ListNet}} = -\frac{1}{B} \sum_{b=1}^B \sum_{k=1}^K P_{b,k}^{\text{gt}} \log(P_{b,k}^{\text{pred}}) \quad (4)$$

where, l is the index of a candidate trajectory, and P_l^{gt} denotes the ground-truth ranking distribution of the candidate trajectories in the human-designed dataset. r_l is the ground-truth ranking of the l th candidate trajectory, and K is the total number of candidate trajectories. P_l^{pred} denotes the probability distribution over candidate trajectories predicted by the reward model, c_l is the score of the l th candidate trajectory, and B represents the batch size.

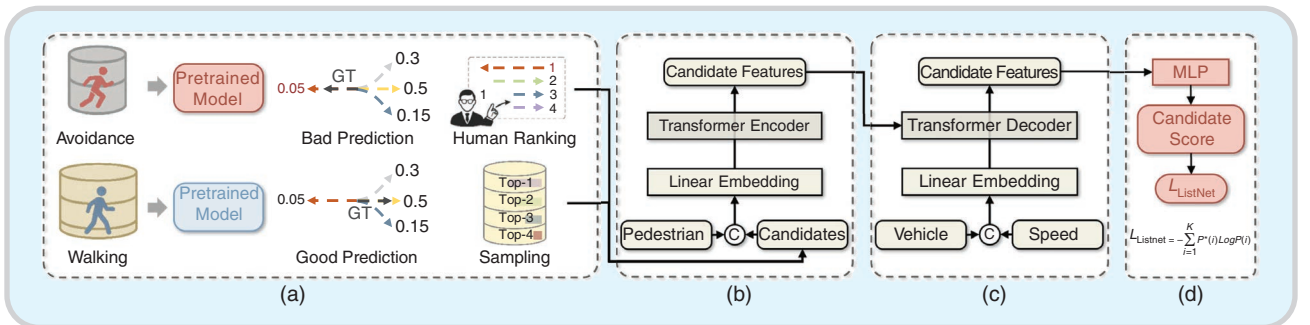


FIG 3 Reward model for reinforcement learning. (a) Human-designed dataset. (b) Candidate trajectory encoder that encodes the human historical trajectory together with the candidate trajectories generated by the pretrained model. (c) Social decoder that takes the encoder outputs together with the historical trajectories and velocities of vehicles as input. (d) Output module consisting of a regression head, which employs the ListNet loss to produce the probabilities of the candidate trajectories. GT: ground truth.

Critic Model

The critic model serves as the value function, providing a low-variance baseline for policy optimization and improving the calibration of learning signals derived from the reward model. Given the observed pedestrian history and the surrounding traffic context, together with a concrete future trajectory sampled from the actor model outputs, the critic model estimates a scalar value. This estimate stabilizes the actor's updates by reducing gradient variance and guiding the policy toward behaviors that are both preference aligned and reliable in safety-critical scenarios. The critic mirrors the actor's backbone to ensure representational consistency. It first forms a state-action sequence by concatenating the pedestrian observation with one sampled future trajectory and projects this sequence through a Transformer encoder. The neighbor embeddings are fused with the encoder outputs via a social Transformer decoder. A value head maps the interaction features to a scalar score for each sampled trajectory. Unlike the actor model, the critic model exposes only this scalar value and is trained solely to provide a stable baseline for PPO.

Reference Model

We maintain a frozen copy of the pretrained model as a fixed-anchor policy. It shares the same structure as the actor model and outputs a reference distribution and predicted trajectories over motion modes. During fine-tuning, we add an explicit penalty between the reference model and the actor model's mode distributions to prevent distributional drift, thereby preserving pretrained competence while allowing calibrated adaptation driven by the reward model.

PPO-Based Fine-Tuning Architecture

The PPO process initializes the actor model from a pretrained checkpoint and keeps a frozen reference model. Each rollout uses an old $\pi_{\theta_{\text{old}}}$ for on-policy sampling. The critic model estimates values, and the reward model supplies scalar rewards. We fine-tune only the actor model's classification head, while all other parameters remain frozen (Figure 4).

Rollout and Buffer Construction

Given an observed pedestrian trajectory, velocity, and hazardous vehicle states, the actor model generates K candidate trajectories $\tau^{(k)}$, each associated with a log-probability. Unlike conventional reinforcement learning where the advantage function is computed along temporal transitions, our task involves evaluating a set of candidate pedestrian trajectories generated by the actor model at each rollout. Therefore, the generalized advantage estimation is formulated over the trajectory index. Given K candidate trajectories $\{\tau^{(1)}, \dots, \tau^{(K)}\}$ sampled from the current policy π_{θ} , the reward model provides normalized scalar feedback r_k for each trajectory.

The critic model estimates its value $V_{\phi}(z, \tau^{(k)})$, and the temporal difference residual for the k th trajectory is computed as

$$\delta_k = r_k + \gamma V_{\phi}(z, \tau^{(k+1)}) - V_{\phi}(z, \tau^{(k)}) \quad (5)$$

where γ is the discount factor. z represents the pedestrian's historical trajectory and the historical trajectories and speed of hazardous vehicles.

Then, the advantage for the k th trajectory is obtained by propagating residuals backward along the trajectory sequence

$$\hat{A}_k = \sum_{l=0}^{K-k} (\gamma \lambda)^l \delta_{k+l} \quad (6)$$

where $\lambda \in (0, 1)$ balances the bias-variance tradeoff.

The empirical return is finally defined as

$$\hat{R}_k = \hat{A}_k + V_{\phi}(\tau^{(k)}). \quad (7)$$

This formulation enables a stable and low-variance advantage estimation in the trajectory-sampling space, ensuring that PPO updates remain effective when training is conducted over discrete candidate trajectories.

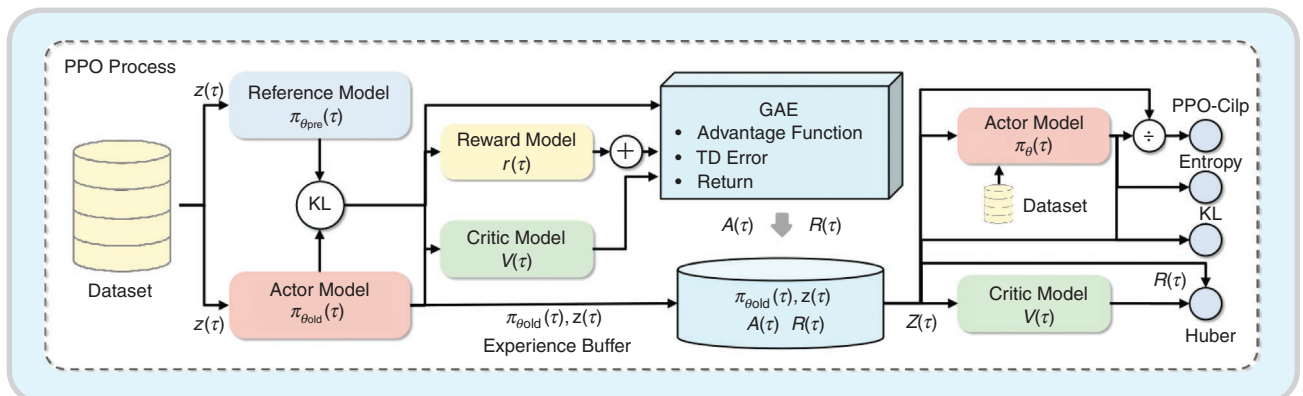


FIG 4 Reinforcement learning from a human-designed reward model based on PPO. It consists of the actor, reference, reward, and critic models [59]. TD: temporal difference.

Policy Update

The PPO objective is constructed by clipping the importance sampling ratio, defined as:

$$\mathcal{L}^{\text{PPO}}(\theta) = \mathbb{E}_k[\min(r_k(\theta)\hat{A}_k, \text{clip}(r_k(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_k)] \quad (8)$$

where $r_k(\theta) = \pi_\theta(\tau^{(k)} | z^k) / \pi_{\theta_{\text{old}}}(\tau^{(k)} | z^k)$, and ϵ is the clipping threshold.

To prevent policy collapse and maintain proximity to the reference model $\pi_{\theta_{\text{ref}}}$, a KL penalty term is added

$$\mathcal{L}^{\text{KL}}(\theta) = D_{\text{KL}}(\pi_\theta | \pi_{\theta_{\text{ref}}}). \quad (9)$$

In addition, a behavior cloning term is used to encourage consistency with ground-truth pedestrian trajectories

$$\mathcal{L}^{\text{BC}}(\theta) = \|\tau_\theta - \tau_{\text{gt}}\|_1 \quad (10)$$

where τ_θ is the predicted trajectory, and τ_{gt} is the ground truth.

Finally, the actor model objective is defined as

$$\mathcal{L}^{\text{Actor}} = -\mathcal{L}^{\text{PPO}} + \beta H(\pi_\theta) + \alpha \mathcal{L}^{\text{KL}} + \lambda \mathcal{L}^{\text{BC}}(\theta) \quad (11)$$

where $H(\pi_\theta)$ denotes entropy regularization to encourage trajectory diversity. α , λ , and β are weighting coefficients.

Value Function Update

The critic model is trained to approximate the value of each candidate trajectory sampled from the actor model. We adopt a clipped regression loss, which has been widely used in PPO to prevent overly large updates of the value function. Formally, the critic loss is defined as

$$\mathcal{L}^{\text{critic}}(\phi) = \frac{1}{2} \mathbb{E}_k[\max((V_\phi(\tau^{(k)}) - \hat{R}_k)^2, (V_\phi^{\text{clip}}(\tau^{(k)}) - \hat{R}_k)^2)] \quad (12)$$

where $V_\phi(\tau^{(k)})$ denotes the current value for the candidate trajectory. $V_\phi^{\text{clip}}(\tau^{(k)})$ is the clipped value, defined as

$$V_\phi^{\text{clip}}(\tau^{(k)}) = V_{\phi_{\text{old}}}(\tau^{(k)}) + \text{clip}(V_\phi(\tau^{(k)}) - V_{\phi_{\text{old}}}(\tau^{(k)}), -\epsilon_v, \epsilon_v) \quad (13)$$

where $V_{\phi_{\text{old}}}$ denotes the critic model before the update and ϵ_v is the clipping threshold.

This design reduces instability and overfitting, which are common issues when training value functions in high-variance settings such as trajectory prediction. The critic model learns to align its value estimates with the empirical returns produced by the reward model and generalized advantage estimation (GAE), thereby providing a stable baseline for advantage calculation in the PPO updates.

Pedestrian Dataset for Safety-Critical Scenarios

A pedestrian trajectory dataset specifically designed for the safety-critical scenarios plays a crucial role in both training and validating the multimodal trajectory prediction model. This dataset provides the foundation for evaluating model performance under real-world hazardous interactions.

VR Experiment Under Safety-Critical Scenarios

Pedestrian precrash behaviors in real traffic are rare, and collecting such data in real traffic is hazardous. To address this gap, we developed high-fidelity safety-critical scenarios using the CARLA simulator (Figure 5). By integrating VR technology with a pedestrian motion simulator, volunteers are able to immerse themselves in interactions with hazardous vehicles. For scenario design, two common accidents from the Future Mobile Traffic Accident Scenario Study Dataset are adopted: jaywalking and unprotected intersection. In-depth accident data indicate that the impact velocity between pedestrians and vehicles in urban areas generally falls within the range of 20–50 km/h. To obtain pedestrian behaviors under different time pressures, the control system positions a hazardous vehicle 18 m away from the pedestrian, moving at constant speeds of 15, 25, 35, and 45 km/h, respectively. Thus, the corresponding time periods were at levels of 4.8, 2.9, 2.1, and 1.6 s, respectively. It is important to note that although the experiment was designed with four time-to-collision levels and fixed approaching positions, the appearance time of the hazardous vehicle is determined randomly based on the pedestrian's position. Specifically, as the width of the hazardous vehicle is 2 m, we define a near interaction zone within 1.5 m of the vehicle's centerline in the crossing direction and a relatively distant interaction zone between 1.5 and 3 m. When the subject reaches a randomly assigned position within either of the aforementioned zones during the crossing task, the hazardous vehicle generation program is activated. Moreover, an obstacle vehicle is also included to block the subject's view, making the scenarios more realistic and hazardous. To ensure randomness, a triggering algorithm is implemented so that hazardous vehicles appear randomly during pedestrian crossings. In these scenarios,

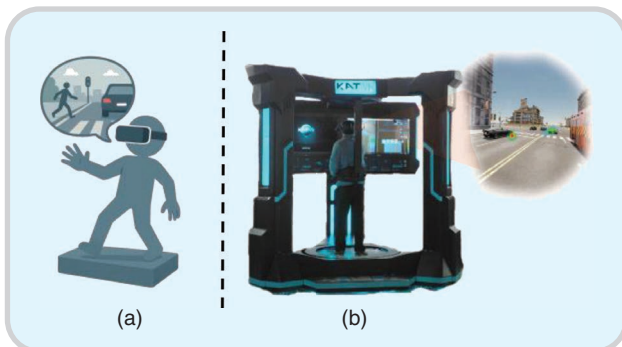


FIG 5 VR experiment with human in the loop under safety-critical scenarios. (a) Scenario diagram and (b) pedestrian simulator.

pedestrians could choose one of four different behaviors: constant speed normal walking, backward, forward accelerating, or stopping. Each experiment captured the interaction process from the onset of pedestrian crossing to the conclusion of vehicle encounter, thereby constructing a comprehensive dataset of safety-critical scenarios.

Dataset

Thirty consent-informed volunteers participated in the experiment. Each volunteer was exposed to 64 distinct scenarios with a 3-min interval between trials to mitigate fatigue. To reduce learning effects, the scenarios were divided into two groups and conducted on different days. During the experiment, volunteers chose the timing of crossing, while hazardous vehicles appeared randomly along their paths. Participants were instructed to respond with behaviors they deemed appropriate. The study protocol was approved by the Institutional Review Board of Tsinghua University. A total of 1,920 interaction segments were collected, including 928 cases of continuous normal walking and 992 involving avoidance. The data are available at <https://github.com/siyuan15-jpg/P-RLHF.git>.

Each segment recorded 1.5–5 s of data from the vehicle's appearance until the pedestrian reached the destination or a collision occurred. It is noteworthy that although the numbers of normal walking and avoidance cases are comparable, avoidance behaviors are brief and embedded within predominantly normal walking. Consequently, for trajectory prediction tasks, the dataset still exhibits a significant imbalance, with common behaviors overshadowing rare ones.

For a more detailed description of the experiment, please refer to our companion article, which provides additional categories of data and an in-depth analysis of pedestrian behaviors in safety-critical scenarios.

Experiment Design and Result Analysis

This section presents the experimental setup and model evaluation. Specifically, we first describe the configurations used for model training. Then, we compare our model with several SOTA approaches for multimodal trajectory prediction. Next, we examine the model's performance before and after fine-tuning on both rare and whole behaviors. The comparison is supported by ablation studies that demonstrate the necessity of the fine-tuning module. Finally, we introduce a human preference evaluation in which volunteers subjectively assessed the performance of the model.

Experimental Setup

Pretrain Model

The observation and prediction horizons are set to four and eight frames, respectively. The time interval between consecutive frames is 0.1 s. The hidden dimension of both the encoder and decoder is set to 128. The model adopts a two-

layer transformer encoder-decoder structure, where each layer contains four attention heads. Eight trajectory clusters are generated by applying k -means clustering on the training data to represent diverse motion modes. The pretrained model is trained for 30 epochs using the Adam optimizer with a learning rate of 1×10^{-4} and a batch size of 128.

Reward Model

The reward model shares the same temporal setup as the pretrained model. The model outputs ranking scores over K candidate trajectories for each pedestrian. It is trained for up to 500 epochs with a batch size of 64, using the Adam optimizer with an initial learning rate of 1×10^{-4} and a weight decay of 1×10^{-5} . To avoid overfitting, early stopping is applied with a patience of 20 epochs.

Actor Model

The actor model is initialized from the pretrained model. During reinforcement learning, only the classification head of the actor model is fine-tuned, while the remaining parameters are frozen. Thirty candidate trajectories are generated per sample during rollouts.

The Adam optimizer is used with a learning rate of 1×10^{-4} . The PPO framework objective incorporates multiple terms: 1) a clipped surrogate objective with a clipping parameter of $\epsilon = 0.2$, 2) an entropy regularization term with a coefficient of 0.2 to encourage exploration, 3) a KL-divergence penalty with a coefficient of 0.02 to constrain deviation from the reference model, and 4) a behavioral cloning loss weighted by one to preserve consistency with ground-truth trajectories. Training proceeds for up to 500 rollouts, with each rollout comprising multiple epochs (four per rollout) and a batch size of 256.

Critic Model

The critic model is designed to estimate the value of predicted trajectories. It is trained jointly with the actor model using the Adam optimizer with a learning rate of 5×10^{-5} . The objective combines a smooth L1 regression loss with a clipped value loss, where the predicted return is constrained within $\epsilon_v = 0.2$ of the target return. Gradient clipping with a maximum norm of 0.5 is applied during optimization to further improve stability.

Quantitative Results

We compared the proposed model with several SOTA approaches on two test sets: the whole- and rare-behavior sets. As many models do not provide probability, we augmented them without altering their backbones, and all the models were retrained on our dataset.

Deep Generative Model

Deep generative models are capable of predicting multiple candidate trajectories, consistent with the stochastic na-

ture of human behavior. We compare four approaches: 1) GPGraph, which learns to assign each pedestrian to the most likely behavior group in an unsupervised manner [63]; 2) SocialVAE, which employs a variational autoencoder with social attention to capture navigation uncertainty [64]; 3) EigenTrajectory, which proposes an anchor-based refinement method built upon an SGCN baseline model to predict pedestrians' latent trajectories [65]; and 4) SingularTrajectory, which adopts a diffusion-based framework to predict pedestrians' multimodal trajectories [40].

Transformer Model

As the Transformer serves as the foundation of our study, we select the TUTR model for comparison [37]. TUTR integrates multimodal prediction and social interaction into a transformer architecture, directly generating diverse trajectories with probabilities. In our pretrained model, only the output position of the classification head is modified, while all other components remain unchanged. In addition, we compare our approach with DSBCL-PRM, a pedestrian trajectory prediction model designed to address the long-tail problem [3].

Performance on Evaluation Metrics

Based on our proposed dataset, we compared the aforementioned models on both the whole-behavior and rare-behavior test sets. The whole-behavior test set contains 6,704 trajectories. Applying the same top-four criterion, we further extract a rare-behavior test set comprising 698 trajectories on which the pretrained model performs poorly. Our model achieved comparable performance in minADE and minFDE, indicating that the pretrained model adapted from TUTR already provides a good baseline. Our primary focus lies on the brierADE and brierFDE metrics as they reflect the model's ability to assign appropriate probabilities to the predicted trajectory. When probability is taken into account, the proposed model outperforms the baseline approaches on most metrics. Compared with the six probability-aware baselines, our model achieves average reductions

of 8.8% in brierADE and 11.2% in brierFDE on the whole-behavior test set, and corresponding average reductions of 1.5% and 5.1% on the rare-behavior test set (Table 1).

Accuracy of the Reward Model

For the eight candidate trajectories generated by the pre-trained model, the reward model assigns scores according to human judgment. We assume that the trajectory closest to the ground truth should be ranked within the top-four predictions, indicating that the model not only captures this motion mode but also assigns it sufficient attention. Accordingly, the reward model is expected to assign a higher probability to the prediction nearest to the ground truth, and we evaluate this capability using top-one and top-four accuracy. The test results show that the reward model achieves 41% top-one accuracy and 80% top-four accuracy, demonstrating satisfactory performance.

Model Comparison Before and After Fine-Tuning

For safety-critical scenarios, microscopic comparisons can better highlight the impact of fine-tuning. We compare the model's performance before and after fine-tuning on both hard-to-predict and easy-to-predict behaviors to show its effectiveness. In addition, we also compared the performance indicators of the model before and after fine-tuning through ablation experiments.

Hard-to-Predict Behaviors

Backward-avoidance and sudden-stop behaviors are hard to predict as pedestrians may exhibit abrupt changes in speed and direction within a short time. Figure 6(a) and (c) illustrates the top-four predicted trajectories before and after fine-tuning, where the pedestrian moves only along the y -axis. Based on the history of normal walking (i.e., blue solid line), the pretrained model predicts continued walking (i.e., green and red dashed lines). However, the pedestrian adopts to retreat (i.e., orange solid line). Although the pretrained model fails to assign sufficient probability to this behavior, the fine-tuned model increases the

Table 1. Performance comparison on the classical trajectory prediction metrics (* represents the conducted model variant).

Category	Method	Venue/Year	minADE (m)	minFDE (m)	b-ADE (m)	b-FDE (m)
Deep generative model	GPGraph*	ECCV2022	0.05 /0.11	0.06 /0.11	0.7/ 0.82	0.74/ 0.87
	SocialVAE*	ECCV2022	0.05 /0.11	0.07/0.15	0.73/0.95	0.75/0.99
	EigenTrajectory*	ICCV2023	0.08/0.11	0.11/0.14	0.7/0.86	0.76/0.9
	SingularTrajectory*	CVPR2024	0.06/0.1	0.08/ 0.1	0.65 /0.86	0.67/ 0.87
Transformer model	TUTR	CVPR2023	0.06/ 0.09	0.09/0.13	0.8/0.9	0.83/0.94
	DSBCL-PRM	TASE2024	0.05 / 0.09	0.06 /0.11	0.7/0.91	0.71/0.93
	Ours	—	0.05 / 0.09	0.06 / 0.1	0.65 /0.87	0.66 / 0.87

The numbers in bold represent the best performance of these seven models on the corresponding metrics.

likelihood of backward avoidance and achieves more accurate prediction. Figure 6(b) and (d) illustrates the model's prediction of sudden-stop behavior. Based on the stationary history, the pretrained model still predicts normal walking, indicating that its output is unreliable in this case. By contrast, the fine-tuned model allocates a reasonable probability to the sudden stop while retaining its ability to predict alternative behaviors.

Easy-to-Predict Behaviors

Normal walking and forward accelerating behaviors are relatively easy to predict, as the pedestrian's direction remains unchanged. Figure 7(a) and (c) shows a normal walking case where the pretrained model assigns nearly uniform probabilities across trajectories, which is unfavorable for downstream safety decision making. By contrast, the fine-tuned model concentrates more probability on the trajectory closest to the ground truth, while also slightly increasing the likelihood of backward avoidance.

For the forward accelerating case [Figure 7(b) and (d)], the concentration of probability around the ground truth remains evident. This indicates that the reward model effectively scores the predictions and discriminates the relative ranking of candidate trajectories, thereby guiding the pretrained model to optimize toward assigning higher probability to the trajectory closest to the ground truth.

Ablation Studies

We evaluate TUTR, the pretrained, and the fine-tuned models on both the whole-behavior and rare-behavior test datasets (Table 2). Ablation results show that, on the whole-behavior dataset, the pretrained model reduces b-ADE by 12.5% and b-FDE by 14.5% compared with TUTR. These gains stem from two specific modifications. First, we reposition the classification head to facilitate subsequent reinforcement learning fine-tuning. Second, we replace the cross-entropy loss used in TUTR with a KL-divergence loss for classification. As a result, the classification head of the pretrained model better fits the probability distribution in the whole-behavior dataset. However, we observe that these improvements do not translate into reduced errors on the rare-behavior test dataset.

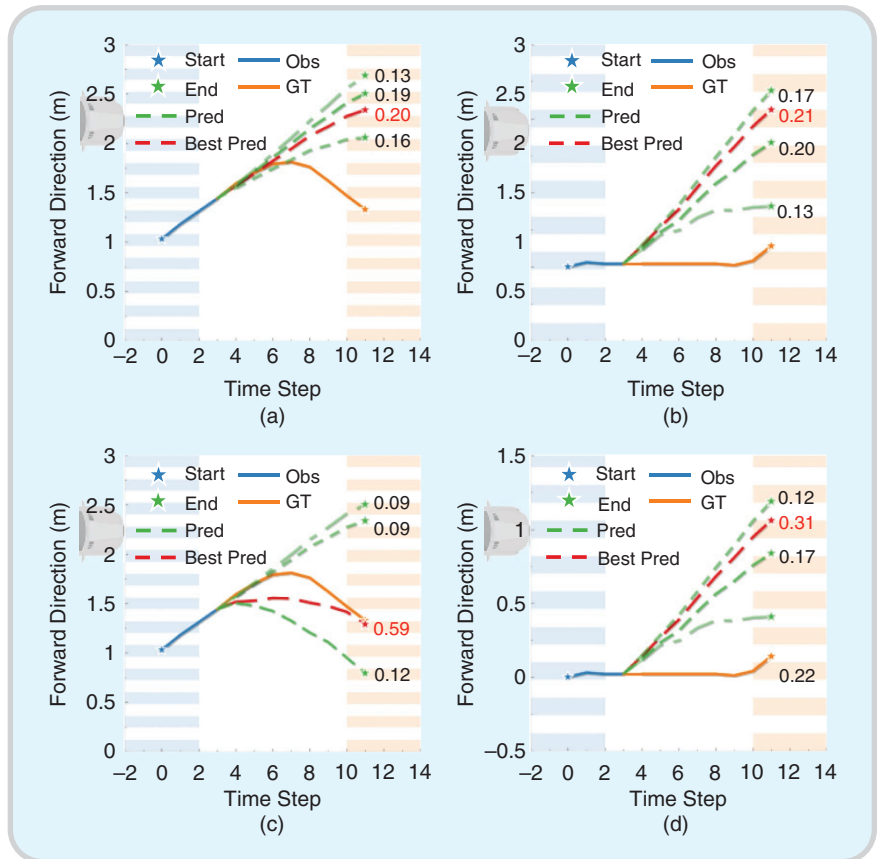


FIG 6 Qualitative comparisons of our model and the pretrained model on hard-to-predict behaviors. Only top-four predicted trajectories ranked by probability are shown for clarity (the red line marks the most probable prediction, and the green lines denote the remainder). (a) Pretrained backward. (b) Pretrained stop. (c) Fine-tuned backward. (d) Fine-tuned stop.

Table 2. Performance comparison among TUTR and the pretrained and fine-tuned models on the classical Trajectory prediction metrics.

Model	Min ADE (m)	Min FDE (m)	b-ADE (m)	b-FDE (m)
TUTR	0.06/0.09	0.09/0.13	0.8/0.9	0.83/0.94
Pre-train	0.05/0.09	0.06/0.1	0.7/0.92	0.71/0.93
Fine-tune	0.05/0.09	0.06/0.1	0.65/0.87	0.66/0.87

Min: minimum.

For models before and after fine-tuning (Table 2), the results show that, after fine-tuning, the model achieves reductions of 7.1% in brierADE and 7% in brierFDE on the whole-behavior set, and reductions of 5.4% and 6.5% on the rare-behavior set. These findings indicate that the proposed model not only maintains accurate predictions for common behaviors but also enhances the prediction of rare behaviors, thereby improving the trajectory prediction task in safety-critical scenarios.

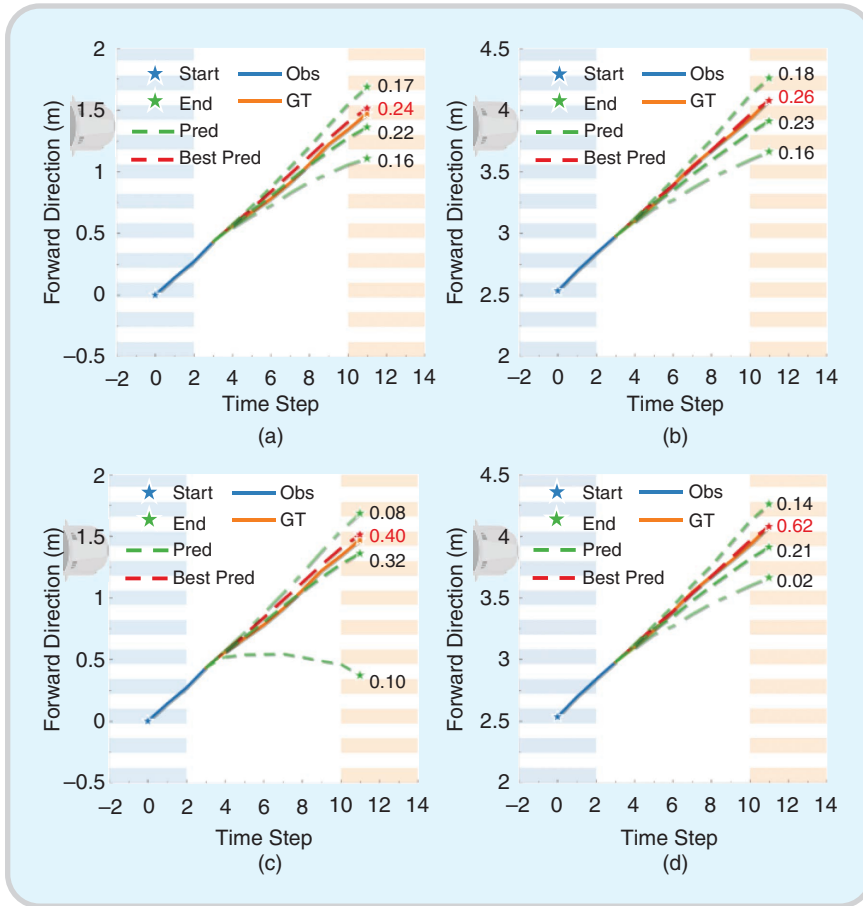


FIG 7 Qualitative comparisons of our model and the pretrained model on easy-to-predict behaviors. Only top-four predicted trajectories ranked by probability are displayed for clarity (the red line marks the most probable prediction, and the green lines denote the remainder). (a) Pretrained forward (walk). (b) Pretrained forward (avoid). (c) Fine-tuned forward (walk). (d) Fine-tuned forward (avoid).

Human Preference Evaluation

Classical metrics are primarily used to evaluate the overall model performance but may fail to properly assess individual cases (see “Details of the Human Preference Evaluation”). Therefore, consistent with related studies, we introduce human preference to subjectively evaluate the model’s performance before and after fine-tuning [14]. Specifically, 400 cases are randomly sampled from the whole-behavior and rare-behavior test sets to form a human evaluation dataset. Visualizations of the model outputs are presented, and seven volunteers, without knowing the correspondence between models and visualizations, are asked to judge each case by selecting one of three options: 1) the pretrained model is better, 2) the fine-tuned model is better, or 3) both are equal. Then, we compute the better-or-equal (BOE) rate as the measure of subjective human judgment

$$\begin{aligned} \text{BOE}_p &= \frac{|\text{Preference} \geq 0|}{200}, \\ \text{BOE}_f &= \frac{|\text{Preference} \leq 0|}{200}, \end{aligned} \quad (14)$$

where BOE_p and BOE_f are the human preference metric of the pretrained and fine-tuned models, respectively. Preference is the number of choices of better or equal.

The results of the human preference evaluation show that, for the whole-behavior evaluation set, the average scores given by seven volunteers are 0.73 for the pretrained model and 0.67 for the fine-tuned model, indicating that the fine-tuned model retains the basic capability of the pretrained one. For the rare-behavior evaluation set, the average scores are 0.47 and 0.88 for the pretrained and fine-tuned models, respectively, with the latter showing a substantial improvement. These results show that the fine-tuned model enhances attention to rare behaviors while maintaining its ability to common behaviors. For detailed information, please refer to “Details of the Human Preference Evaluation.”

Limitation

This study has several limitations. First, the pedestrian trajectory prediction model considers only basic information such as trajectory and velocity, without incorporating map information. Second, the interaction scenarios collected in VR still differ from real-world conditions as only perpendicular pedestrian–vehicle interactions are modeled, while cases such as rear-end and turning scenarios remain to be included. Third, the construction of the reward model training dataset involves a degree of subjectivity, and defining the dataset appropriately is critical for effective RLHF fine-tuning.

Conclusion

This study proposed a trajectory prediction model fine-tuned with a human-designed reward model to address the challenge of long-tail trajectory prediction in safety-critical scenarios. By integrating a Transformer-based trajectory predictor with reinforcement learning fine-tuning, the proposed model maintains accuracy on common behaviors while enhancing its focus on rare behaviors. Experimental results showed that, on our pedestrian dataset, the model achieves a better performance on brierADE and brierFDE. This approach provided a solution to the bias caused by imbalanced data distributions, enabling the model to better adapt to extreme scenarios. However, it must be said that the improvement in rare behavior comes

Details of the Human Preference Evaluation

The purpose of the human preference evaluation is to collect subjective judgment of model outputs before and after fine-tuning. Volunteers were instructed to compare two models based on the positions and probabilities of their predicted trajectories. Figure S1 provides a visualization example. During the evaluation, volunteers were blindfolded so that they could not see the correspondence between figures and models. As a result, some volunteers preferred Figure S1(a) because the probability of the prediction closest to the ground truth is higher than in Figure S1(b). Others judged the two models as comparable because the summed probabilities of the top-three closest predictions were similar. Conversely, some volunteers favored Figure S1(b), noting that Figure S1(a) assigns overly low probabilities to its top-three predictions. All subjective judgments were carefully recorded, and a better-or-equal score was computed for each volunteer.

A t-test was conducted to examine whether fine-tuning yields statistically significant differences on the whole-behavior and rare-behavior datasets (Table S1). For the whole-behavior dataset, the t-statistic is 2.92 with a p value of 0.03, indicating no statistically significant difference between the pretrained and fine-tuned models at the 0.01 significance level. By contrast, for the rare-behavior dataset, the t-statistic is -24.9 with a p value far below 0.01, demonstrating a statistically significant performance improvement after fine-tuning.

Table S1. Performance comparison between the pretrained and fine-tuned models on the human preference metric.

Subject	Whole Behavior		Rare Behavior	
	Pretrained	Fine-tuned	Pretrained	Fine-tuned
001	0.73	0.63	0.42	0.86
002	0.98	0.95	0.51	0.93
003	0.61	0.59	0.42	0.9
004	0.81	0.75	0.51	0.86
005	0.73	0.65	0.48	0.88
006	0.58	0.58	0.46	0.86
007	0.7	0.52	0.5	0.87
Average	0.73	0.67	0.47	0.88

These results indicate that fine-tuning preserves performance on common behaviors while achieving substantial and statistically significant gains on rare behaviors.

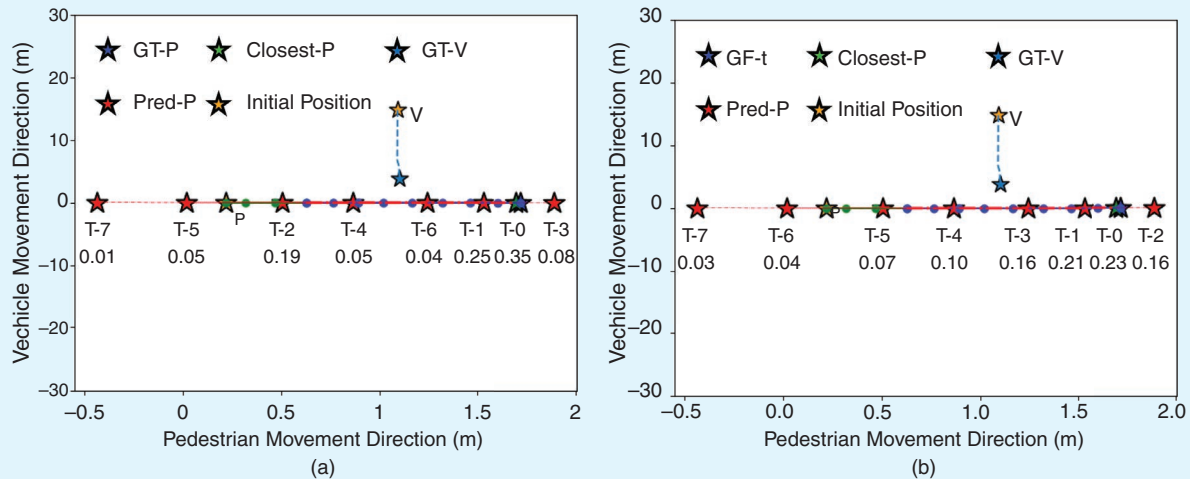


FIG S1 Human preference evaluation visualization. (a) Fine-tuned model. (b) Pretrained model.

at the expense of reduced accuracy in common behavior, and how to balance these two remains an open challenge.

Acknowledgment

This work was supported by National Natural Science Foundation of China under Grant 52572413 and Grant 52402450. This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Institutional Review Board of Tsinghua University.

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